

Carl Friedrich Gauß Werke — Band 6

Carl Friedrich Gauß

8.

Ad maiorem illustrationem praeceptorum praecedentium, calculum integrum, per quem systema elementorum tertium ex oppositionibus annorum 1805, 1807, 1808, 1809 determinatum est, hic apponam. Supposui longitudinem nodi ascendentis pro initio anni 1803, $172^{\circ} 28' 46''.8 = \beta$, inclinationem orbitae $34^{\circ} 37' 31''.5$; reduxi illam ad tempora singularem oppositionum, addendo praecessionem $2' 26''.01$, $3' 37''.50$, $4' 39''.12$, $5' 37''.11$. Hinc derivavi ex longitudinibus latitudines heliocentricas

$$\begin{aligned}g &= -33^{\circ} 40' 50''.63 \\g' &= +28^{\circ} 14' 51''.24 \\g'' &= +27^{\circ} 20' 55''.86 \\g''' &= -4^{\circ} 52' 28''.44\end{aligned}$$

Porro prodeunt coefficientes

$$\begin{aligned}c &= +0.1252, & f &= -0.9870 \\c' &= -0.3366, & f' &= +0.8917 \\c'' &= +0.3609, & f'' &= +0.8727 \\c''' &= +0.6803, & f''' &= -0.1811\end{aligned}$$

Porro prodeunt argumenta latitudinum

$$\begin{aligned}257^{\circ} 25' 6''.73 \\56^{\circ} 24' 5''.96 \\126^{\circ} 2' 55''.07 \\188^{\circ} 36' 2''.69\end{aligned}$$

e quibus elementa elliptica elicere oportet. Ad hunc finem statuo angulum $\varphi = 14^{\circ} 10' 4''.08$ (uti in systemate elementorum secundo inventus erat), longitudinemque perihelii, pro initio anni 1803, $= 121^{\circ} 5' 22''.1$. Perihelium (perinde ut nodus) respectu stellarum fixarum immobile supponitur, adeoque ipsius distantia a nodo ascendente constans $= 308^{\circ} 36' 35''.3$. Hinc habemus per formulas notas:

$$\begin{aligned}v &= 308^{\circ} 47' 30''.66, & M &= 327^{\circ} 54' 28''.87, \\v' &= 107^{\circ} 55' 26''.71, & M' &= 79^{\circ} 49' 5''.20, \\v'' &= 177^{\circ} 15' 13''.11, & M'' &= 175^{\circ} 54' 19''.77, \\v''' &= 239^{\circ} 59' 27''.39, & M''' &= 266^{\circ} 29' 57''.59\end{aligned}$$

Porro

$$\begin{aligned} m &= -0.68517, & n &= +1.18569 \\ m' &= -1.06482, & n' &= -2.01318 \\ m'' &= -1.59701, & n'' &= -0.12921 \\ m''' &= -1.18352, & n''' &= +1.93947 \end{aligned}$$

Denotando igitur correctionem longitudinis perihelii per $d\varpi$, atque correctionem anguli φ per $d\varphi$, erit motus medius sidereus

ab oppositione prima usque ad secundam intra dies 521.140706

$$111^{\circ} 30' 41''.97 - 0.37965 d\varpi - 3.19887 d\varphi$$

ab oppositione secunda usque ad tertiam intra dies 449.277674

$$96^{\circ} 8' 1''.82 - 0.53219 d\varpi + 1.88397 d\varphi$$

ab oppositione tertia usque ad quartam intra dies 422.786667

$$90^{\circ} 35' 28''.72 + 0.41849 d\varpi + 2.06868 d\varphi$$

Hinc eruuntur tres expressiones pro valore medio diurno sidereo

$$\begin{aligned} 770''.31398 - 0.0007285 d\varpi - 0.0061382 d\varphi \\ 770''.30718 - 0.0011845 d\varpi + 0.0041933 d\varphi \\ 771''.37892 + 0.0009780 d\varpi + 0.0048930 d\varphi \end{aligned}$$

e quarum comparatione demanant aequationes:

$$\begin{aligned} 0 &= 0''.00680 + 0.0004560 d\varpi - 0.0103315 d\varphi \\ 0 &= -1''.07174 - 0.0021625 d\varpi - 0.0006997 d\varphi \end{aligned}$$

atque hinc $d\varpi = -488''.82$, $d\varphi = -20''.92$, ac proin

longitudo perihelii correcta pro initio anni 1803, $120^{\circ} 57' 13''.28$,
valorque correctus anguli $\varphi = 14^{\circ} 9' 43''.16$

Porro fit motus medius diurnus sidereus $770''.7985$

anomaliam media correcta pro oppositione prima $328^{\circ} 20' 55''.20$

longitudo media pro eadem epocha $89^{\circ} 20' 34''.49$

Denique derivatur ex motu medio sidereo logarithmus semiaxis maioris 0.4420439, adeoque logarithmus semiparametri 0.4152361. Iam quum habeamus valores correctos anomaliarum verarum

$$\begin{aligned} v &= 308^{\circ} 56' 40''.25 \\ v' &= 107^{\circ} 55' 39''.48 \\ v'' &= 177^{\circ} 34' 28''.59 \\ v''' &= 240^{\circ} 7' 36''.21 \end{aligned}$$

computantur logarithmi radiorum vectorum

$$\begin{aligned}\log r &= 0.3531088 \\ \log r' &= 0.4492406 \\ \log r'' &= 0.5369700 \\ \log r''' &= 0.4716739\end{aligned}$$

E tabulis solaribus porro habemus

$$\begin{aligned}\log R &= 9.9937332 \\ \log R' &= 0.0039862 \\ \log R'' &= 0.0065917 \\ \log R''' &= 0.0011160\end{aligned}$$

unde tandem deducimus

$$\begin{aligned}\gamma &= -33^{\circ} 39' 48''.15 \\ \gamma' &= +28^{\circ} 15' 0''.73 \\ \gamma'' &= +27^{\circ} 20' 9''.07 \\ \gamma''' &= -4^{\circ} 52' 53''.99\end{aligned}$$

Ut eruantur coefficientes a , b etc., formo hypothesin secundam, retinendo inclinationem sed augendo longitudinem nodi uno minuto primo, ut sit pro initio anni 1803, $172^{\circ} 29' 46''.8$. Statuendo dein, ut in hypothesi prima, longitudinem perihelii pro eadem epocha = $121^{\circ} 5' 22''.1$, inveniuntur anomaliae verae

$$\begin{aligned}v &= 308^{\circ} 48' 40''.93 \\ v' &= 107^{\circ} 47' 34''.06 \\ v'' &= 177^{\circ} 26' 22''.25 \\ v''' &= 239^{\circ} 59' 15''.00\end{aligned}$$

atque hinc (statuendo quoque $\varphi = 14^{\circ} 10' 4''.08$) anomaliae mediae

$$\begin{aligned}M &= 328^{\circ} 15' 51''.59, \\ M' &= 79^{\circ} 46' 30''.67, \\ M'' &= 175^{\circ} 54' 32''.83, \\ M''' &= 266^{\circ} 29' 42''.93\end{aligned}$$

(Ceterum ad hunc computum non opus est, methodum vulgarem adhibere, sed sufficit, valoribus anomaliarum mediarum in hypothesi prima inventis addere producta e coefficientibus m , m' , m'' , m''' positive sumtis in differentias inter valores respectivos anomaliarum verarum in utraque hypothesi, puta

$$\begin{aligned}+9''.50 \times 0.68517 &= +6''.51 \\ +3''.40 \times 1.06482 &= +3''.62 \\ +2''.48 \times 1.59701 &= +3''.96 \\ -12''.39 \times 1.18352 &= -14''.66\end{aligned}$$

Hinc retinendo valores coefficientium m , n etc. eruitur

$$\begin{aligned}
d\varpi &= -468''.21, \text{ longitudo perihelii initio anni 1803, } 120^{\circ} 57' 33''.89 \\
d\varphi &= -20''.62, \varphi = 14^{\circ} 9' 43''.46 \\
\text{motus medius diurnus sidereus} &= 770''.7761 \\
\text{logarithmus semiaxis maioris} &= 0.4420523 \\
\text{longitudo media in oppositione prima} &= 89^{\circ} 20' 47''.85
\end{aligned}$$

$$\begin{aligned}
\gamma &= -33^{\circ} 39' 51''.10 \\
\gamma' &= +28^{\circ} 15' 1''.27 \\
\gamma'' &= +27^{\circ} 20' 10''.97 \\
\gamma''' &= -4^{\circ} 52' 54''.66
\end{aligned}$$

Tandem formo hypothesin tertiam statuendo $\beta = 172^{\circ} 28' 46''.6$, $i = 34^{\circ} 38' 31''.5$, unde perinde ut ante deducitur

$$\begin{aligned}
\text{longitudo perihelii pro initio anni 1803, } &120^{\circ} 55' 34''.46 \\
\varphi &= 14^{\circ} 9' 52''.63 \\
\text{motus medius diurnus sidereus} &= 770''.8398 \\
\text{logarithmus semiaxis maioris} &= 0.4420283 \\
\text{longitudo media in oppositione prima} &= 89^{\circ} 20' 20''.65
\end{aligned}$$

$$\begin{aligned}
\gamma &= -33^{\circ} 39' 35''.63 \\
\gamma' &= +28^{\circ} 15' 5''.20 \\
\gamma'' &= +27^{\circ} 20' 9''.32 \\
\gamma''' &= -4^{\circ} 52' 52''.65
\end{aligned}$$

Comparatio valorum latitudinum heliocentricarum $\gamma, \gamma', \gamma'', \gamma'''$ in tribus hypothesibus inventarum producit

$$\begin{aligned}
a &= -0.0492, & b &= +0.2087 \\
a' &= +0.0212, & b' &= +0.0745 \\
a'' &= +0.0317, & b'' &= +0.0042 \\
a''' &= -0.0112, & b''' &= +0.0223
\end{aligned}$$

Habentur itaque quatuor aequationes

$$\begin{aligned}
+62''.48 - 0.1744 d\beta + 1.1957 di &= 0 \\
+9''.49 + 0.3578 d\beta - 0.8172 di &= 0 \\
-46''.79 - 0.3292 d\beta - 0.8685 di &= 0 \\
-25''.55 - 0.6915 d\beta + 0.2034 di &= 0
\end{aligned}$$

e quibus per principium quadratorum minimorum deducitur

$$\begin{aligned}
d\beta &= -54''.41 \\
di &= -42''.06
\end{aligned}$$

ita ut habeatur

longitudo nodi ascendentis pro initio anni 1803, 172ř 27' 52".39
inclinatio orbitae 34ř 36' 49".44

Elementa reliqua vel per comparisonem eorum, quae in singulis tribus hypothesibus prodierunt, erui, vel quod accuratius est, per calculum novum longitudinum in orbita atque repetitionem operationum in art. 5 explicatarum determinari poterunt. Methodus prima suppeditat

longitudinem perihelii 1803	120ř 58' 3".86
angulum φ	14ř 9' 36".25
longitudinem mediam in oppositione prima	89ř 20' 32".08
motum medium diurnum sidereum	770".7899
logarithmum semiaxis maioris	0.4420471

Per methodum alteram invenitur

longitudo perihelii 1803	120ř 58' 4".81
angulus φ	14ř 9' 36".63
longitudo media in oppositione prima	89ř 20' 31".81
motus medius diurnus sidereus	770".7893
logarithmus semiaxis maioris	0.4420473

Cum his elementis conveniunt ea, quae supra (art. 3) tradidi.

9.

Quantumvis magnae sint perturbationes, quas Pallas a reliquis planetis patitur, tamen, experientia teste, elementa elliptica quatuor oppositionibus adaptata planetae motui intra integrum hoc tempus satis bene satisfaciunt: quin adeo etiam a motu antecedente ac consequente, nisi intervallum temporis nimium assumatur, parum differunt, ita ut e. g. elementa secunda in art. 3 tradita, in oppositione anni 1803 tribus, in oppositione anni 1809 quinque minutis primis a longitudine heliocentrica observata discrepaverint. Quae quum ita sint, ad construendam ephemeridem pro motu planetae futuro semper commodissimum mihi videtur elementis pure ellipticis uti, quae ex quatuor oppositionibus proxime antecedentibus derivata erant, siquidem multitudo aequationum a perturbationibus oriundarum tanta certo evasura est, ut computus vel unius loci heliocentrii planetae tantum non aequè operosus evadere debeat, ac calculus elementorum ellipticorum per praecepta supra tradita. Ita locum planetae geocentricum salvo errore paucorum minorum primorum tuto semper praedicere licebit, quae praecisio ad inveniendum planetam sufficit.

10.

Attamen postulat scientiae dignitas, ut consensui stabiliiori prospiciatur, quem antequam perturbationes in calculum introducantur obtineri non posse manifestum est. Praematurus fuisset, meo quidem iudicio, calculus tam prolixus taedique plenus, quamdiu observationum copia tempus nimis exiguum complectebatur, perturbationesque a planetis reliquis oriundae vix se manifestabant. Nunc vero, ubi motus ellipticus non amplius sufficit, ad omnia loca observata inter se concilianda, tempus adesse videtur, ubi de theoria accuratiori cogitari potest. Quo pacto calculum perturbationum, quas Pallas

patitur praesertim a Iove, commodissime atque exactissime absolvendum censeam, quum methodis pro aliis planetis adhibitis propter nimiam excentricitatem atque inclinationem vix ac ne vix quidem uti liceat, mox alio loco fusius explicabo: de elementis autem ellipticis, quae maxime idonea videantur, ut calculus perturbationum ipsis superstruatur, in sequentibus adhuc agam. Eruenda scilicet mihi propono elementa elliptica, quae non his vel illis oppositionibus exacte, sed omnibus, quae hactenus observatae sunt, quam proxime satisfaciant. Methodum quidem, per quam tale negotium exsequi liceat, iam in Theoria Motus Corporum Coelestium art. 187 succincte descripsi: sed quum non solum ea, quae illic generaliter tractavi, in casu speciali, ubi loca observata sunt oppositiones, quaedam compendia admittant, sed etiam quaedam artificia practica, per quae applicationem methodi quadratorum minimorum faciliorem reddere iamdudum solitus sum, in opere illo desiderentur, astronomis haud ingratum fore spero, si hosce calculos aliquanto fusius hie tradidero. Quum totum negotium versetur in determinatione correctionum elementis approximatis, quae ab omnibus locis observatis haud multum aberrare supponuntur, adiiciendarum, totus labor a duobus momentis pendebit: primo scilicet formari debent aequationes lineares, quas singula loca observata suppeditant, dein ex his aequationibus valores incognitarum maxime idonei sunt eruendi.

11.

Sit secundum elementa approximata

L	longitudo media planetae pro epocha arbitraria
t	numerus dierum inde ab epocha usque ad momentum observationis elapsorum
γ	motus medius diurnus sidereus in minutis secundis
ϖ	longitudo perihelii
$e = \sin \varphi$	excentricitas
a	semiaxis maior
r	radius vector
v	anomaliam vera
E	anomaliam excentricam
β	longitudo nodi ascendentis
i	inclinatio orbitae
u	argumentum latitudinis
λ	longitudo heliocentrica
Y	latitudo heliocentrica
δ	latitudo geocentrica
R	distantia terrae a Sole

Ex observatione autem suppono esse

α	longitudinem heliocentricam
δ	latitudinem geocentricam

Denique per dL , $d\gamma$, $d\varpi$ etc. denoto correctiones quantitatum L , γ , ϖ etc.

Fit itaque $dL + t d\gamma$ correctio longitudinis mediae,

$dL + t d\gamma - d\varpi$ correctio anomaliae mediae,

adeoque per art. 15, 16 Theoriae Motus Corp. Coel.

$$dE = \frac{1}{\cos \varphi} (dL + t d\gamma - d\varpi) + \frac{\tan \varphi (2 - e \cos E - e^2 \sin^2 E)}{\cos \varphi} d\varphi$$

$$dr = -da + a \tan \varphi \sin E (dL + t d\gamma - d\varpi) - a \cos \varphi \cos v d\varphi$$

Porro fit correctio argumenti latitudinis $du = d\varpi + d\beta - d\beta$,
atque per art. 52 Theoriae Motus C. C. correctio longitudinis heliocentricae:

$$d\lambda = d\beta - \tan Y \cos(\lambda - \beta) di + \frac{r}{\cos Y} du$$

Hinc colligitur

$$\begin{aligned} d\lambda &= \frac{r}{a \cos Y} dL \\ &+ \frac{rt \cos \varphi \sin E}{a \cos Y} d\gamma \\ &+ \left(\frac{r}{a \cos Y} - \frac{\tan Y \cos(\lambda - \beta)}{r \cos Y} \right) d\varpi \\ &- \tan Y \cos(\lambda - \beta) di \\ &+ d\beta + \frac{r}{a \cos Y} (2 - e \cos E - e^2) \sin E d\varphi \end{aligned}$$

Porro quum habeatur

$$\begin{aligned} a^3 \gamma^2 &= \text{Const.} \\ r \sin(\delta - \gamma) &= R \sin \delta \\ \tan \gamma &= \tan i \sin(\alpha - \beta) \end{aligned}$$

fit per differentiationem

$$\begin{aligned} \frac{da}{a} + \frac{2 d\gamma}{3\gamma} &= 0 \\ \frac{dR}{R} + \cot(\delta - \gamma) \cdot (d\delta - d\gamma) &= \cot \delta d\delta \end{aligned}$$

sive

$$\begin{aligned} \frac{dR}{R} &= \frac{\sin \beta \cos(\beta - \gamma)}{\sin \gamma} d\delta \\ \frac{d\gamma}{\sin^2 \gamma} &= \frac{d\alpha}{\sin^2 i} - \sin 2\gamma \cot(\alpha - \beta) d\beta \end{aligned}$$

unde adhibito valore ipsius dr supra evoluto colligitur

$$\begin{aligned}
d\delta = & -\frac{a \sin \beta \sin(\beta - \gamma)}{r \sin \gamma} da \\
& + \frac{a \sin \beta \sin(\beta - \gamma) \tan \varphi \sin E}{r \sin \gamma} (dL + t d\gamma - d\varpi) \\
& - \frac{a \sin \beta \sin(\beta - \gamma) \cos \varphi \cos v}{r \sin \gamma} d\varphi \\
& + \frac{a \sin \beta \sin(\beta - \gamma)}{r \sin \gamma} (2 - e \cos E - e^2) \sin E d\varphi \\
& - \frac{\sin \beta \cos(\beta - \gamma)}{\sin \gamma} d\delta \\
& - \frac{a \sin \beta \sin(\beta - \gamma)}{r \sin \gamma} d\delta
\end{aligned}$$

Hinc valores longitudinis heliocentricae atque latitudinis geocentricae fiunt ex valoribus correctis elementorum $\lambda + d\lambda$, $\delta + d\delta$, adeoque quaevis oppositio sup- peditat binas aequationes:

$$\begin{aligned}
\alpha &= \lambda + d\lambda \\
\delta &= \delta + d\delta
\end{aligned}$$

12.

Applicando haecce praecepta ad sex oppositiones Palladis in art. 2 traditas, si calculum secundo elementorum systemati in art. 3 delineato superstruimus, sequentes duodecim aequationes obtinemus:

Ex oppositione prima, ubi longitudo computata inventa est = $277^{\circ} 36' 20''.07$, latitudo geocentrica = $+4^{\circ} 26' 29''.19$:

$$\begin{aligned}
0 &= -183''.93 + 0.79363 dL + 143.66 d\gamma + 0.39493 d\varpi + 0.95920 d\varphi \\
&\quad - 0.18856 d\beta + 0.17387 di \\
0 &= -6''.81 - 0.02658 dL + 46.71 d\gamma + 0.02658 d\varpi - 0.20858 d\varphi \\
&\quad + 0.15946 d\beta + 1.25782 di
\end{aligned}$$

Ex oppositione secunda, ubi longitudo computata = $337^{\circ} 0' 36''.04$, latitudo geocentrica = $+15^{\circ} 1' 46''.71$:

$$\begin{aligned}
0 &= -0''.06 + 0.58880 dL + 358.12 d\gamma + 0.26208 d\varpi - 0.85234 d\varphi \\
&\quad + 0.14912 d\beta + 0.17775 di \\
0 &= -3''.09 + 0.01318 dL + 28.39 d\gamma - 0.01318 d\varpi - 0.07861 d\varphi \\
&\quad + 0.91704 d\beta + 0.54365 di
\end{aligned}$$

Ex oppositione tertia, ubi longitudo computata = $67^{\circ} 20' 42''.88$, latitudo geocentrica = $-54^{\circ} 31' 3''.88$:

$$\begin{aligned}
0 &= +0''.02 + 1.73436 dL + 1846.17 d\gamma - 0.54603 d\varpi - 2.05662 d\varphi \\
&\quad - 0.18833 d\beta - 0.17445 di \\
0 &= -8''.98 - 0.12606 dL - 227.42 d\gamma + 0.12606 d\varpi - 0.38939 d\varphi \\
&\quad + 0.17176 d\beta - 1.35441 di
\end{aligned}$$

Ex oppositione quarta, ubi longitudo computata = $223^{\circ} 37' 25''.39$, latitudo geocentrica = $+42^{\circ} 11' 28''.07$:

$$\begin{aligned} 0 &= -2''.31 + 0.99584 dL + 1579.03 d\gamma + 0.06456 d\varpi + 1.99545 d\varphi \\ &\quad - 0.06040 d\beta - 0.33750 di \\ 0 &= +2''.47 - 0.08089 dL - 67.22 d\gamma + 0.08089 d\varpi - 0.09970 d\varphi \\ &\quad - 0.46359 d\beta + 1.22803 di \end{aligned}$$

Ex oppositione quinta, ubi longitudo computata = $304^{\circ} 2' 59''.71$, latitudo geocentrica = $+37^{\circ} 44' 31''.82$:

$$\begin{aligned} 0 &= +0''.01 + 0.65311 dL + 1329.09 d\gamma + 0.38994 d\varpi - 0.08439 d\varphi \\ &\quad - 0.04305 d\beta + 0.34268 di \\ 0 &= +38''.12 - 0.00218 dL + 38.47 d\gamma + 0.00218 d\varpi - 0.18710 d\varphi \\ &\quad + 0.47301 d\beta - 1.14371 di \end{aligned}$$

Ex oppositione sexta, ubi longitudo computata = $359^{\circ} 34' 46''.67$, latitudo geocentrica = $-7^{\circ} 20' 12''.13$:

$$\begin{aligned} 0 &= -317''.73 + 0.69957 dL + 1719.32 d\gamma + 0.12913 d\varpi - 1.38787 d\varphi \\ &\quad + 0.17130 d\beta - 0.08360 di \\ 0 &= +117''.97 - 0.01315 dL - 43.84 d\gamma + 0.01315 d\varpi + 0.02929 d\varphi \\ &\quad + 1.02138 d\beta - 0.27187 di \end{aligned}$$

Sed ex his duodecim aequationibus decimam omnino reiiciemus, quum latitudo geocentrica observata nimis incerta sit.

13.

Quum sex incognitas dL , $d\gamma$ etc. ita determinare non liceat, ut omnibus undecim aequationibus exacte satisfiat, i. e. ut singulae incognitarum functiones quae sunt ad dextram simul fiant = 0, valores eos eruemus, per quos functionum harum quadrata summam quam minimam efficiant. Facile quidem perspicitur, si generaliter functiones lineares incognitarum p , q , r , s etc. propositae sint hae

$$\begin{aligned} n + ap + bq + cr + ds + \text{etc.} &= w \\ n' + a'p + b'q + c'r + d's + \text{etc.} &= w' \\ n'' + a''p + b''q + c''r + d''s + \text{etc.} &= w'' \\ n''' + a'''p + b'''q + c'''r + d'''s + \text{etc.} &= w''' \end{aligned}$$

etc., aequationes conditionales, ut $ww + w'w' + w''w'' + w'''w''' + \text{etc.} = \Omega$ fiat minimum, esse hasce

$$\begin{aligned} aw + a'w' + a''w'' + a'''w''' + \text{etc.} &= 0 \\ bw + b'w' + b''w'' + b'''w''' + \text{etc.} &= 0 \\ cw + c'w' + c''w'' + c'''w''' + \text{etc.} &= 0 \\ dw + d'w' + d''w'' + d'''w''' + \text{etc.} &= 0 \end{aligned}$$

etc. sive, designando brevitatis causa

$$\begin{aligned} an + a'n' + a''n'' + a'''n''' + \text{etc.} &= [an] \\ aa + a'a' + a''a'' + a'''a''' + \text{etc.} &= [aa] \\ ab + a'b' + a''b'' + a'''b''' + \text{etc.} &= [ab] \quad \text{etc.} \end{aligned}$$

$$\begin{aligned} bb + b'b' + b''b'' + b'''b''' + \text{etc.} &= [bb] \\ bc + b'c' + b''c'' + b'''c''' + \text{etc.} &= [bc] \quad \text{etc. etc.} \end{aligned}$$

per

$$nn + n'n' + n''n'' + n'''n''' + \text{etc.} = [nn]$$

p, q, r, s etc. determinari debere per eliminationem ex aequationibus

$$\begin{aligned} [an] + [aa]p + [ab]q + [ac]r + [ad]s + \text{etc.} &= 0 \\ [bn] + [ab]p + [bb]q + [bc]r + [bd]s + \text{etc.} &= 0 \\ [cn] + [ac]p + [bc]q + [cc]r + [cd]s + \text{etc.} &= 0 \\ [dn] + [ad]p + [bd]q + [cd]r + [dd]s + \text{etc.} &= 0 \quad \text{etc.} \end{aligned}$$

Attamen quoties multitudo incognitarum p, q, r, s etc. paullo maior est, eliminatio laborem valde prolixum atque taediosum requirit, quem sequenti modo notabiliter contrahere licet. Praeter coefficientes $[an], [aa], [ab]$ etc. (quorum multitudo fit $\frac{1}{4}(n^2 + 3n)$, si multitudo incognitarum $= n$), etiam hunc computatum suppono $nn + n'n' + n''n'' + n'''n''' + \text{etc.} = [nn]$, perspicieturque facile fieri

$$\begin{aligned} \Omega &= [nn] + 2[an]p + 2[bn]q + 2[cn]r + 2[dn]s + \text{etc.} \\ &+ [aa]pp + 2[ab]pq + 2[ac]pr + 2[ad]ps + \text{etc.} \\ &+ [bb]qq + 2[bc]qr + 2[bd]qs + \text{etc.} \\ &+ [cc]rr + 2[cd]rs + \text{etc.} \\ &+ [dd]ss + \text{etc.} \quad \text{etc.} \end{aligned}$$

Designando itaque

$$[an] + [aa]p + [ab]q + [ac]r + [ad]s + \text{etc.} = A$$

patet, singulas partes ipsius $\frac{\partial \Omega}{\partial p}$, quae factorem p involvunt, in Ω contineri, adeoque $\Omega - \frac{A^2}{[aa]}$ esse functionem a p liberam. Quare statuendo

$$\begin{aligned} [bn] - \frac{[ab]}{[aa]}[an] &= [bn, 1] \\ [cn] - \frac{[ac]}{[aa]}[an] &= [cn, 1] \quad \text{etc.} \end{aligned}$$

$$\begin{aligned} [bb] - \frac{[ab]^2}{[aa]} &= [bb, 1] \\ [bc] - \frac{[ab][ac]}{[aa]} &= [bc, 1] \quad \text{etc. etc.} \end{aligned}$$

erit

$$\begin{aligned}\Omega - \frac{A^2}{[aa]} &= [nn, 1] + 2[bn, 1]q + 2[cn, 1]r + 2[dn, 1]s + \text{etc.} \\ &+ [bb, 1]qq + 2[bc, 1]qr + 2[bd, 1]qs + \text{etc.} \\ &+ [cc, 1]rr + 2[cd, 1]rs + \text{etc.} \\ &+ [dd, 1]ss + \text{etc.} \quad \text{etc.}\end{aligned}$$

quam functionem designabimus per Ω' .

14.

Simili modo ponendo

$$[bn, 1] + [bb, 1]q + [bc, 1]r + [bd, 1]s + \text{etc.} = B$$

erit $\Omega' - \frac{B^2}{[bb, 1]}$ functio a q libera, quam statuemus $= \Omega''$. Eodem modo faciemus

$$[cn, 1] - \frac{[bc, 1]}{[bb, 1]}[bn, 1] = [cn, 2] \quad \text{etc.}$$

$$[cc, 1] - \frac{[bc, 1]^2}{[bb, 1]} = [cc, 2] \quad \text{etc.}$$

etc. etc. atque

$$[cn, 2] + [cc, 2]r + [cd, 2]s + \text{etc.} = C$$

unde $\Omega'' - \frac{C^2}{[cc, 2]}$ functio ab r quoque libera. Eodem modo progrediemur, usquedum in progressionem $\Omega, \Omega', \Omega''$ etc. ad terminum ab omnibus incognitis liberum pervenerimus, qui erit $[nn, n]$, si multitudo incognitarum p, q, r, s etc. denotatur per n . Habemus itaque

$$\Omega = \frac{A^2}{[aa]} + \frac{B^2}{[bb, 1]} + \frac{C^2}{[cc, 2]} + \frac{D^2}{[dd, 3]} + \text{etc.} + [nn, n]$$

Iam quum $\Omega = ww + w'w' + w''w'' + \text{etc.}$ natura sua valorem negativum obtinere non possit, facile demonstratur, divisores $[aa], [bb, 1], [cc, 2], [dd, 3]$ etc. necessario positivos evadere debere (brevitatis tamen gratia hanc demonstrationem fusi-
us hic non exsequor). Hinc vero sponte sequitur, valorem minimum ipsius Ω prodire, si fiat $A = 0, B = 0, C = 0, D = 0$ etc. Ex his itaque n aequationibus incognitae p, q, r, s etc. determinari debebunt, quod ordine inverso facillime effici potest, quum manifesto ultima aequatio unicam incognitam implicet, penultima duas et sic porro. Simul haec methodus eo nomine se commendat, quod valor minimus aggregati Ω sponte inde innotescit, quippe qui manifesto est $= [nn, n]$.

15.

Applicemus iam haecce praecepta ad exemplum nostrum, ubi p, q, r, s etc. sunt $dL, d\gamma, d\varpi, d\varphi, d\beta, di$. Calculo accurate absoluto hosce valores numericos inveni:

$[nn] = 148848$	$[cn, 1] = -119.31$	$[df, 2] = -0.17265$
	$[ee, 2] = +2.24325$	
$[an] = -371.09$	$[dn, 1] = -125.18$	
$[bn] = -580104$	$[en, 1] = +72.52$	$[ef, 2] = -0.37841$
$[cn] = -113.45$	$[fn, 1] = -43.22$	$[ff, 2] = +5.61686$
$[dn] = +268.53$	$[bb, 1] = +2458225$	
$[en] = +94.26$	$[bc, 1] = +62.13$	
$[nn, 2] = +99034$		
$[fn] = -31.81$	$[bd, 1] = -510.58$	$[dn, 2] = +25.66$
$[aa] = +5.91569$	$[be, 1] = +213.84$	$[en, 2] = +74.23$
$[ab] = +7203.91$	$[bf, 1] = +73.45$	
$[ac] = -0.09344$	$[cc, 1] = +0.71769$	$[fn, 2] = +2.75$
$[ad] = -2.28516$	$[cd, 1] = +1.09773$	$[dd, 2] = +9.29213$
$[ae] = -0.34664$	$[ce, 1] = -0.05852$	$[de, 2] = -0.36175$
$[af] = -0.18194$	$[cf, 1] = +0.26054$	$[df, 2] = -0.57384$
$[bb] = +10834225$	$[dd, 1] = +11.12064$	$[ee, 3] = +2.23754$
$[bc] = -49.06$	$[de, 1] = -0.50528$	$[ef, 3] = -0.35532$
$[bd] = -3229.77$	$[df, 1] = -0.18790$	$[ff, 3] = +5.52342$
$[be] = -198.64$	$[ee, 1] = +2.26185$	
$[bf] = -143.05$	$[ef, 1] = -0.37202$	$[nn, 4] = +98963$
$[cc] = +0.71917$	$[ff, 1] = +5.61905$	$[en, 4] = +75.23$
$[cd] = +1.13382$		$[fn, 4] = +4.33$
$[ce] = +0.06400$	$[nn, 2] = +117763$	$[ee, 4] = +2.22346$
$[cf] = +0.26341$	$[cn, 2] = -115.81$	$[ef, 4] = -0.37766$
$[dd] = +12.00340$	$[dn, 2] = -153.95$	$[ff, 4] = +5.48798$
$[de] = -0.37137$	$[en, 2] = +84.57$	
$[df] = -0.11762$	$[fn, 2] = -39.08$	
$[ee] = +2.28215$	$[cc, 2] = +0.71612$	$[nn, 5] = +96418$
$[ef] = -0.36136$	$[cd, 2] = +1.11063$	$[fn, 5] = +17.11$
$[ff] = +5.62456$	$[ce, 2] = -0.06392$	$[ff, 5] = +5.42383$
	$[cf, 2] = +0.25868$	
	$[dd, 2] = +11.01463$	$[nn, 6] = +96364$
	$[de, 2] = -0.46088$	
		$[ff, 6] = +5.41853$

Habemus itaque ad determinationem incognitarum sex aequationes sequentes:

$$0 = +17''.11 + 5.42383 di$$

$$0 = +75''.23 + 2.22346 d\varphi - 0.37766 di$$

$$0 = +25''.66 + 9.29213 d\varpi - 0.36175 d\varphi - 0.57384 di$$

$$0 = -115''.81 + 0.71612 dn + 1.11063 d\varphi - 0.06392 d\beta + 0.25868 di$$

$$0 = -138534 + 2458225 d\gamma + 62.13 dn - 510.58 d\varphi + 213.84 d\beta \\ + 73.45 di$$

$$0 = -371''.09 + 5.91569 dL + 7203.91 d\gamma - 0.09344 dn - 2.28516 d\varpi \\ - 0.34664 d\varphi - 0.18194 di$$

unde deducitur

$$\begin{aligned}
 di &= -3''.15 \\
 d\varphi &= -34''.37 \\
 d\varpi &= -4''.29 \\
 dn &= +166''.44 \\
 d\gamma &= +0''.054335 \\
 dL &= -3''.06
 \end{aligned}$$

Elementa itaque elliptica correcta, quae omnibus sex oppositionibus quam proxime satisfaciunt, haec sunt:

Epocha longitudinis mediae 1803, ad meridianum Gottingensem	221 ^h 34' 53''.64
Motus medius tropicus diurnus	770''.5010
Longitudo perihelii 1803	121 ^h 8' 8''.54
Longitudo nodi ascendentis 1803	172 ^h 28' 12''.43
Inclinatio orbitae	34 ^h 37' 28''.35
Excentricitas = $\sin 14^{\circ} 9' 59''.79$	0.2447424
Logarithmus semiaxis maioris	0.4422071

16.

Substitutis valoribus correctionum dL , $d\gamma$ etc., quos modo invenimus, in duodecim aequationibus art. 12, differentias sequentes inter valores longitudinum heliocentricarum atque latitudinum geocentricarum observatos atque computatos obtinemus:

oppositione anni	Differentia	
	longitudinis	latitudinis
1803	-68''.6	-371''.00
1804	-19.92	+0.07
1805	+59.18	-19.92
1807	+85.77	+25.01
1808	+216.54	—
1809	+135.88	-2883.70